AZAZEL audio

VISUALCOMP 2.25

THE COMPRESSOR YOU CAN SEE

USER MANUAL / VERSION 2.25 / VST3 / STANDALONE / WINDOWS & macOS

AZAZEL AUDIO / DESIGNED AND BUILT BY ARNAV SINGH

00 Contents

QUICK START

Insert VisualComp 2.25 on a bus, leave the default Mastering Glue preset loaded, and play your material. Lower **THRESHOLD** until the needle rests around −2 to −4 dB on peaks. That is a finished mix–bus compressor setting – everything else in this manual is refinement.

TWO THINGS WORTH KNOWING IMMEDIATELY

Drag anywhere inside a control's box to change it – you do not have to hit the knob itself. **Hold Ctrl** (Cmd on macOS) while dragging for fine adjustment, and **double–click** any control to reset it.

01 Welcome & Design Philosophy

Compression is the most powerful process in mixing and the hardest to learn, because with most compressors you cannot see what is happening. You turn a knob, something changes – maybe – and it takes years to build the ear that connects a number to a result.

VisualComp 2.25 is built on one idea: if you can see compression working, you can master it. Two oscilloscope displays show your signal before and after processing, with the threshold drawn straight across the waveform. Set the attack too slow and you will *see* the transient escape. Set the release too fast and you will *see* the pumping.

WHAT VISUALCOMP 2.25 IS FOR

- **Glue** – bonding a mix or a bus into one coherent, finished–sounding whole.
- **Control** – evening out an inconsistent vocal, bass or drum performance.
- **Character** – four modelled detector circuits, each with its own timing behaviour and saturation, from clean VCA to warm variable–mu tube.
- **Rhythm** – external sidechain ducking with a visual overlay, so a kick can carve space in a bass line and you can see precisely how deep the duck goes.
- **Learning** – an honest, real–time picture of what dynamics processing does.

SIGNAL FLOW

Input Gain → detector (internal or external sidechain) → gain reduction with your chosen circuit → dry/wet Mix → Auto Gain → Output Gain → optional 0 dB ceiling limiter. Metering and both waveform displays are tapped after the wet/dry blend, so what you see is what leaves the plugin.

> **A NOTE ON THE INTERFACE**
>
> Everything is drawn as resolution–independent vector graphics – no bitmaps – so the panel stays crisp at every zoom setting from 50 % to 200 %, on any display.

02 Installation

WINDOWS – INSTALLER

Open the `Windows` folder and run Install VisualComp 2.25.bat. Windows will ask for administrator permission, which the installer needs in order to write to the shared VST3 folder. It installs:

- **the VST3 plugin into** `C:\Program Files\Common Files\VST3\`
- **the standalone application into** `C:\Program Files\Azazel Audio\VisualComp 2\`
- **a Start Menu shortcut for the standalone**

Then open your DAW and rescan plugins.
`FL Studio:` **Options → Manage plugins → Find more plugins.**
`Ableton / Cubase / Studio One / REAPER:` **rescan in preferences.**

The plugin appears as VisualComp 2, manufacturer Azazel Audio, category Dynamics.

Manual install

If you prefer not to run the installer, copy the `VisualComp 2.25.vst3` folder from the `Windows` folder into `C:\Program Files\Common Files\VST3\` yourself, and run `VisualComp 2.25.exe` from wherever you like.

> **IF THE PLUGIN DOES NOT APPEAR**
> **Custom plugin search paths in FL Studio apply to VST2 only. VST3 plugins are loaded exclusively from the standard VST3 folder above – placing the file elsewhere will not work. Also close and reopen any plugin instance after updating: DAWs keep the old binary in memory while it is loaded.**

02 Installation (continued)

MACOS

macOS plugins must be compiled on a Mac. The `Mac` folder contains the complete source and a one–command build script.

1. Copy the `Mac` folder to your Mac.
2. Install the prerequisites, if you do not already have them:
   ```
   xcode-select --install
   brew install cmake
   ```
3. In Terminal:
   ```
   cd Mac/Source/build-macos
   chmod +x build.sh && ./build.sh
   ```
4. The script builds a universal binary (Apple Silicon + Intel, macOS 10.13+) and installs it to `~/Library/Audio/Plug-Ins/VST3/`. Rescan plugins in your DAW.

The plugin is not code–signed. If your DAW refuses to load it, clear the quarantine flag:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/VST3/"VisualComp 2.25.vst3"
```

WHERE YOUR FILES LIVE

ITEM	WINDOWS	MACOS
Plugin	`C:\Program Files\Common Files\VST3\`	`~/Library/Audio/Plug-Ins/VST3/`
Standalone	`C:\Program Files\Azazel Audio\VisualComp 2\`	(built alongside the plugin)
User presets	`Documents\Azazel Audio\VisualComp 2\Presets\`	`~/Documents/Azazel Audio/VisualComp 2/Presets/`

03 The Guided Tour

The first time you open VisualComp 2.25, an eight–step tour walks you through the panel. Each step dims the interface, spotlights one area and explains it in a sentence or two.

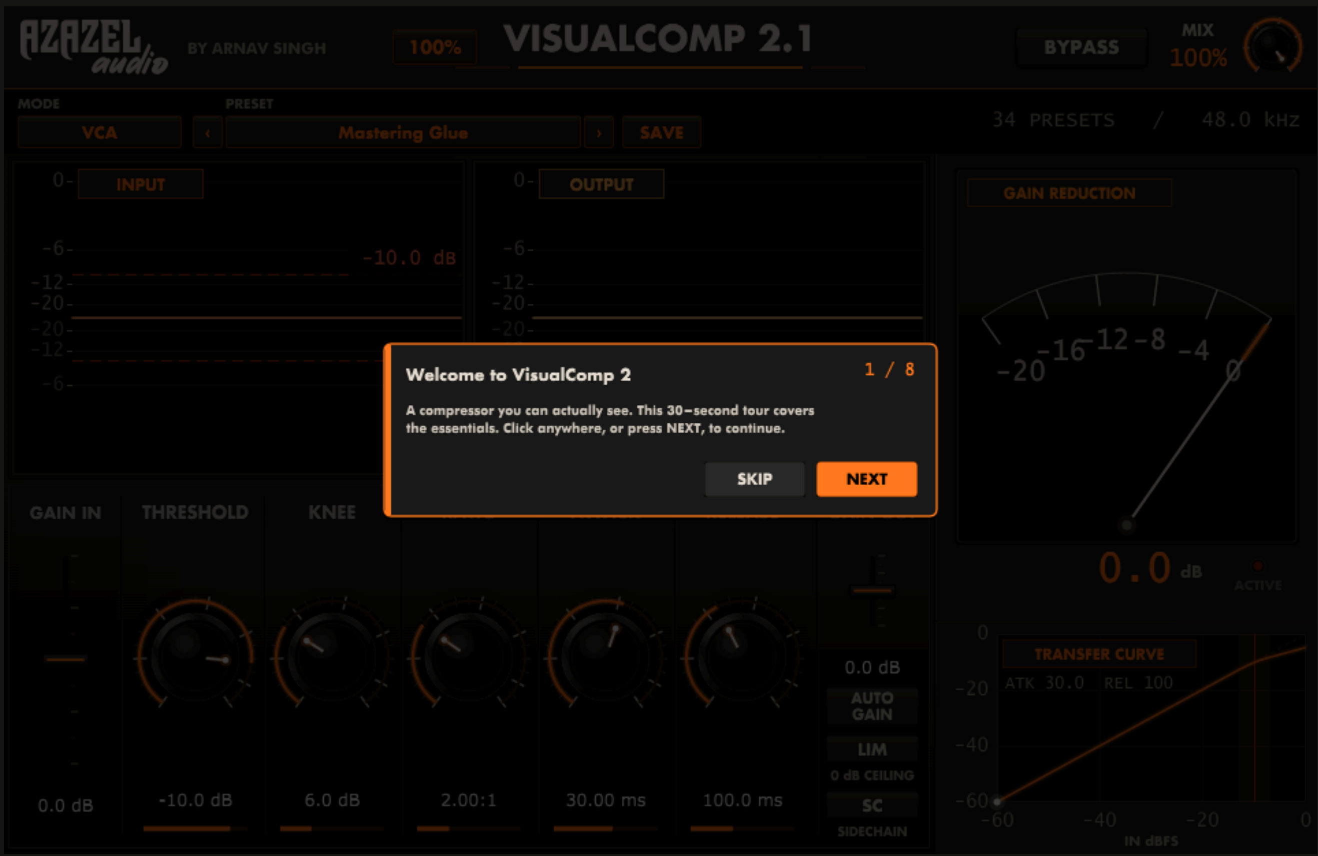

Figure 3.1 – Step one. **NEXT** (orange) advances, **SKIP** (charcoal) dismisses the tour for good. Clicking anywhere on the dimmed area also advances.

Figure 3.2 – Steps spotlight the area they describe. Here the dynamics row is highlighted while the tour explains full-box dragging and the Ctrl fine-adjust modifier.

BRINGING THE TOUR BACK

If you skip the tour it will not appear again. To see it any time, **click the Azazel Audio logo** in the top-left corner. The logo menu offers **Show Help**, which restarts the tour from step one, and **Visit azazelaudio.com**. The "seen" flag is stored per user, so the tour appears exactly once no matter how many instances you load.

WHAT THE TOUR COVERS

#	STEP	IN SHORT
1	Welcome	What the plugin is and how the tour works
2	Presets	Browse, step through with the arrows, save your own
3	Circuit	VCA, FET, Optical and Tube in one line each
4	Displays	Input vs output, and the dashed threshold line
5	Controls	Drag anywhere in a box; Ctrl for fine; double–click to reset
6	Meter	Reading gain reduction, and how much to aim for
7	Sidechain	Routing the key signal, including the FL Studio steps
8	Help	Where to find the tour and the website again

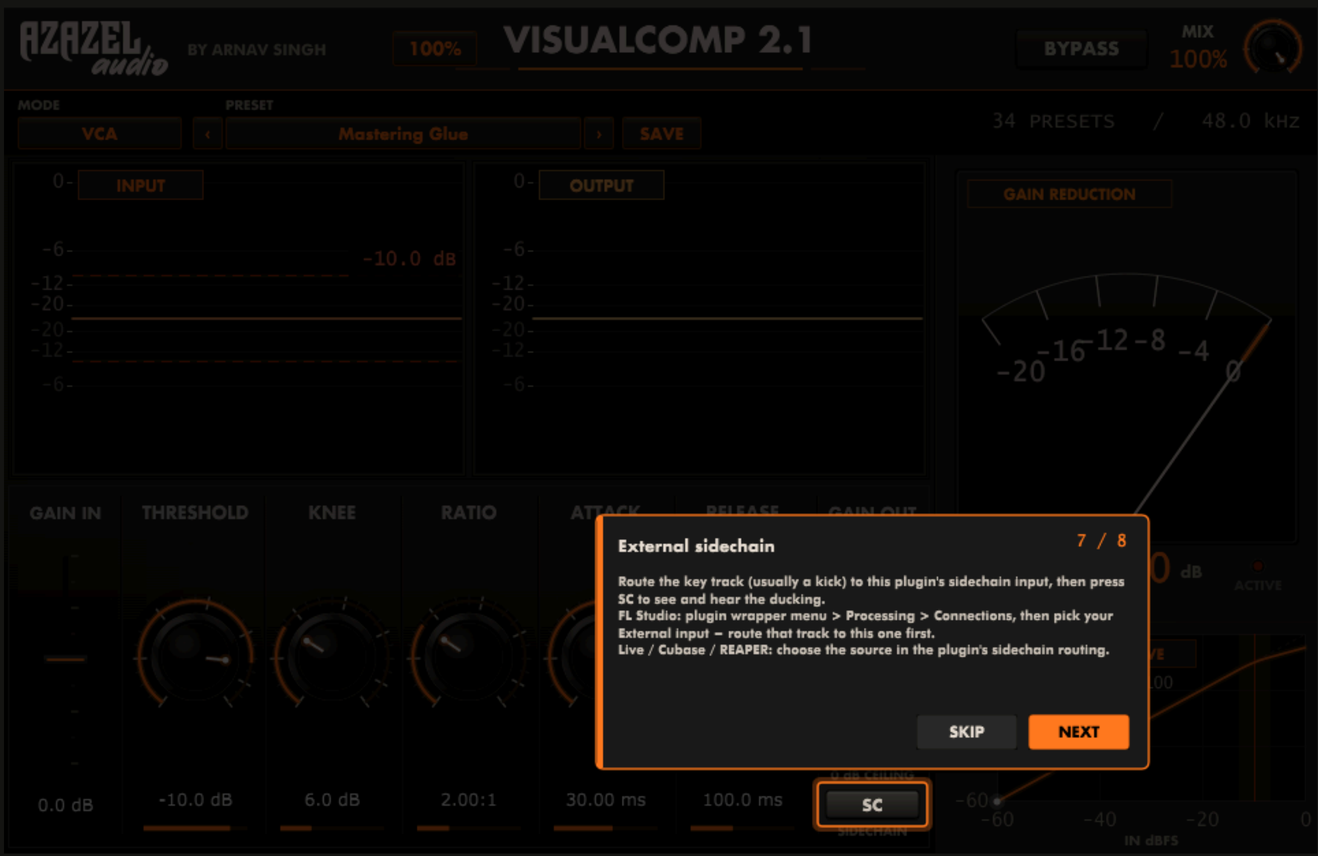

Figure 3.3 – The sidechain step includes the exact FL Studio routing path, since that is where most people get stuck. Section 10 covers every host in detail.

04 Interface Map

Figure 4.1 – The complete VisualComp 2.25 panel at 100 % zoom, showing the default **Mastering Glue** preset.

1. **Azazel logo** – click for Show Help and azazelaudio.com.
2. **Zoom selector** – 50 %, 70 %, 100 %, 150 %, 200 %.
3. **Mode selector** – the detector circuit: VCA, FET, Optical or Tube.
4. **Preset browser** – click the name to browse; arrows step; SAVE stores your own.
5. **Bypass** – lights red and passes audio through untouched.
6. **Mix** – dry/wet blend for parallel compression, with a live percentage.
7. **Input display** – incoming signal with the dashed red threshold line and, when SC is on, the sidechain ducking overlay.
8. **Output display** – the processed result, for direct visual comparison.
9. **Gain In fader** – drive into the detector, O’24 dB.
10. **Threshold, Knee, Ratio, Attack, Release** – the five core controls, each with a graduated ring, an orange value arc and a position bar beneath its read–out.
11. **Gain Out fader** – makeup and final level, O’24 dB.
12. **Auto Gain** – automatic loudness–matched makeup gain.
13. **LIM** – brickwall 0 dB output ceiling.
14. **SC** – enables the external sidechain detector and its overlay.
15. **Gain reduction meter** – analogue needle, 0 to –20 dB, with an activity lamp.
16. **Transfer curve** – the in/out characteristic with a live operating–point dot.

05 The Header & Preset Strip

Figure 5.1 – Header and preset strip. Left to right: logo, author credit, zoom, product name, bypass and mix. The lower strip carries the mode selector and preset browser, with the preset count and current sample rate on the right.

MODE SELECTOR

Click to choose the detector circuit. Each entry names its typical use. The choice is saved with your project and with presets. See section 09.

PRESET BROWSER

- **Click the name field** – opens the full library, organised into category submenus, with your own presets listed below.
- **‹ and ›** – step to the previous or next factory preset without opening a menu. Ideal for auditioning quickly while audio plays.
- **SAVE** – writes the current settings to your personal preset folder.

MIX

Blends the compressed signal against the untouched input. At **100 %** you hear only compressed signal. Lowering it produces parallel compression: heavy compression underneath the natural dynamics, which raises the quiet detail without flattening the peaks.

BYPASS

Passes audio through untouched and resets metering. The button lights red so the panel state is unambiguous at a glance.

ZOOM

Scales the entire interface. The setting is remembered per instance and restored with your project.

ZOOM	PANEL SIZE	SUITED TO
50 %	480 × 310	Small laptop screens, tight layouts
70 %	672 × 434	1080p with several plugins open
100 %	960 × 620	Default – designed size
150 %	1440 × 930	1440p and 4K displays
200 %	1920 × 1240	4K, or detailed waveform work

06 Using the Controls

Every control in VisualComp 2.25 responds the same way, and the rules are worth learning once because they make the plugin considerably faster to operate than a conventional interface.

DRAG ANYWHERE IN THE BOX

Each parameter owns the whole rectangular area around it – the label, the empty space and the value read–out are all part of the same target. Click and drag **anywhere inside a parameter's box** and it responds. You never have to aim precisely at the knob, which matters at small zoom levels and on a trackpad.

Drag **up** to increase, **down** to decrease.

HOLD CTRL FOR FINE ADJUSTMENT

Hold **Ctrl** (**Cmd** on macOS) before you begin dragging and the control becomes roughly eight times less sensitive. This is how you set an attack of 12.4 ms rather than "somewhere near twelve", or trim a threshold by a few tenths of a decibel.

DOUBLE–CLICK TO RESET

Double–clicking any knob or fader returns it to its default value. Useful when an experiment has gone somewhere strange and you want a known starting point without reloading the preset.

GESTURE	RESULT
Drag anywhere in the box	Adjusts that parameter
Ctrl (Cmd) + drag	Fine adjustment, ~8ă more precise
Double–click	Resets the parameter to its default
Mouse wheel over a control	Steps the value up or down

READING THE POSITION BARS

Beneath each of the five main read–outs is a thin orange bar showing where that control sits in its full range. It gives you an instant read of the whole panel's state from across the room – useful when comparing presets, since the bar pattern is recognisable at a glance.

> **AUTOMATION**
> Every parameter is exposed to your DAW for automation, and the panel follows host automation in real time – including the position bars and the transfer curve.

07 The Waveform Displays

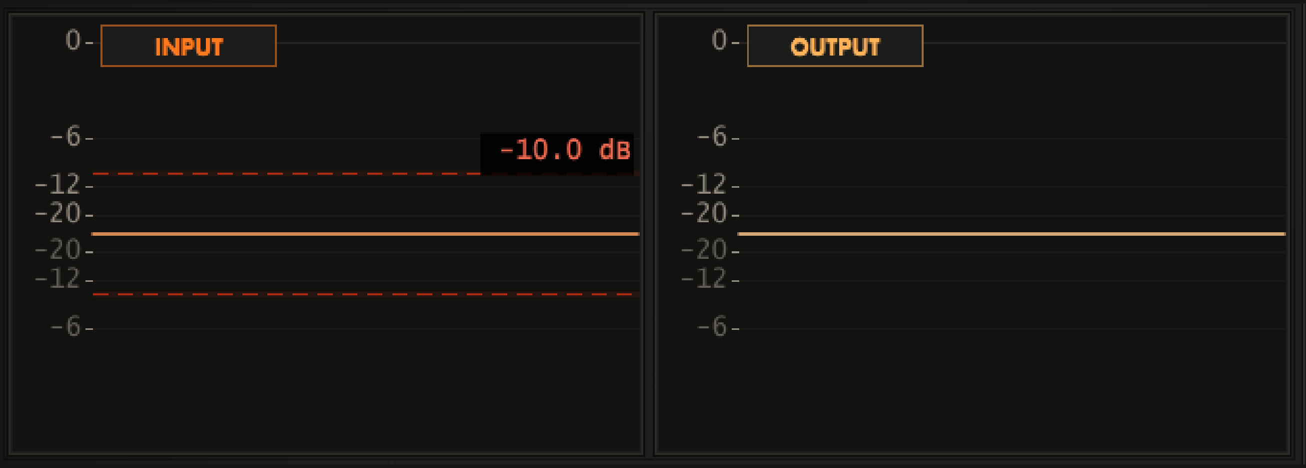

Figure 7.1 – Input (left) and output (right) displays. Each shows about 1.3 seconds of scrolling audio with a mirrored dB scale. The dashed red line across the input is the threshold; its exact value is printed in the chip at the right–hand end.

READING THE DISPLAYS

- New audio enters at the **right** edge and scrolls left, like a DAW playhead view.
- The vertical scale is marked at **0, −6, −12** and **−20 dBFS**, mirrored above and below the centre line.
- Each pixel column holds the true minimum and maximum of the samples it represents, so short transients are never hidden by averaging.
- The **dashed red threshold line** appears on the input display only. Anything crossing it is being compressed.

HOW TO USE THEM

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
Peaks poking well above the dashed line	Working range is healthy	Nothing – normal operation
Nothing reaches the dashed line	No compression at all	Lower THRESHOLD or raise GAIN IN
Output looks like a flat sausage	Over–compression; dynamics gone	Raise THRESHOLD, lower RATIO, or reduce MIX
Sharp spikes survive on the output	Attack too slow to catch them	Shorten ATTACK, or engage LIM as a safety net
Output envelope visibly wobbling	Release chasing the material	Lengthen RELEASE

Tip: toggle **BYPASS** while watching the output display. The difference in envelope shape between the two states is the clearest possible picture of what your settings are doing.

08 The Dynamics Section

Figure 8.1 – The dynamics row. Gain In on the left, the five core controls in the centre, Gain Out with Auto Gain, LIM and SC on the right.

THRESHOLD

`-60 to 0 dB / default -10 dB`

The level above which compression begins. Signal below it passes untouched; signal above it is reduced according to the ratio. This is your primary control over how much compression happens – watch the dashed line move across the input display as you turn it.

RATIO

`1:1 to 20:1 / default 2:1`

How firmly signal above the threshold is reduced. At 4:1, four decibels of input above the threshold become one decibel of output.

RATIO	CHARACTER	TYPICAL USE
1.5:1 – 2:1	Gentle levelling, nearly transparent	Mastering, mix bus, glue
3:1 – 4:1	Clear, musical control	Vocals, drum bus, guitars
6:1 – 10:1	Firm, obvious compression	Bass, rap vocals, taming peaks
12:1 – 20:1	Limiting territory	Parallel crush, safety, effects

KNEE

`0 to 20 dB / default 6 dB`

The width of the transition into compression. At **0 dB** (hard knee) compression begins abruptly at the threshold – precise and obvious. Widening the knee makes the ratio increase gradually either side of the threshold, so the onset is smooth and difficult to hear. Wide knees suit vocals, buses and mastering; hard knees suit drums and deliberate pumping.

08 The Dynamics Section (continued)

ATTACK

`0.1 to 200 ms / default 30 ms`

How quickly gain reduction is applied once the threshold is crossed. Short attacks catch transients and sound controlled but can dull the punch of a drum hit; longer attacks let the initial transient through before clamping down, which is what makes a compressed drum bus sound punchier rather than softer.

- `0.1 – 3 ms` – catches everything. Peak control, limiting, aggressive effect.
- `5 – 15 ms` – general purpose. Vocals, guitars, snare body.
- `20 – 40 ms` – transient–preserving. Drum bus, mix bus, mastering.
- `50 – 200 ms` – slow and gentle. Level–riding under a performance.

RELEASE

`1 to 2000 ms / default 100 ms`

How quickly gain returns once the signal falls back below the threshold. Release is what makes compression feel rhythmic: timed to recover just before the next beat, the compressor breathes with the track. Too fast and you get audible pumping or distortion on bass material; too slow and quiet passages stay suppressed.

- `1 – 50 ms` – very fast. Aggressive, pumping, effect–oriented.
- `80 – 200 ms` – musical default for most sources and buses.
- `300 – 600 ms` – smooth level–riding for vocals and bass.
- `800 – 2000 ms` – very slow, effectively an automatic fader.

> **THE RELIABLE ORDER OF OPERATIONS**
>
> Set **RATIO** for the character you want, lower **THRESHOLD** until the needle shows the gain reduction you want, then shape **ATTACK** and **RELEASE** while watching the output display. Finally match levels with **GAIN OUT** or **AUTO GAIN** and A/B against bypass.

09 Detector Circuits

The mode selector changes the compressor's internal behaviour, not merely a label. Each circuit imposes its own timing constraints on your knob settings and contributes its own saturation character, so the same numbers genuinely sound different in different modes.

MODE	BEHAVIOUR	TIMING CHARACTER	BEST ON
VCA	Clean, symmetrical, precise. Follows your settings exactly with no added colour.	Honours attack and release as set	Mix bus, drum bus, anything needing control without character
FET	Very fast detector with musical saturation that adds bite and forward presence.	Attack clamped to 2 ms or faster; release at least 20 ms	Drums, bass, rap and rock vocals, parallel crush
Optical	Photocell–style gain element with programme–dependent recovery. Leans on the signal rather than grabbing it.	Attack at least 30 ms; release at least 100 ms	Lead vocals, acoustic guitar, bass, anything that must sound effortless
Tube	Variable–mu behaviour: the effective ratio deepens as the signal drives harder, plus gentle harmonic warmth.	Attack at least 50 ms; release at least 200 ms	Lead vocals, mix bus, mastering, warmth and weight

WHY YOUR KNOB VALUES MAY NOT APPLY LITERALLY

Real optical and tube compressors physically cannot react in a fraction of a millisecond, and a FET unit cannot be slow. VisualComp 2.25 respects those limits: in Optical and Tube modes very short attack settings are lengthened to the circuit's minimum, and in FET mode the attack is always fast. The knobs remain fully useful – they simply operate within each circuit's real range.

CHOOSING A CIRCUIT QUICKLY

- Need it invisible and accurate? VCA.
- Need it punchy and forward? FET.
- Need it smooth and forgiving? Optical.
- Need it warm and weighty? Tube.

10 Sidechain – Routing in Your DAW

External sidechaining lets one signal control the compression applied to another – most commonly a kick drum ducking a bass line or pad so the low end stays clear. VisualComp 2.25 exposes a second stereo input bus named Sidechain for exactly this.

> **THE RULE THAT TRIPS EVERYONE UP**
>
> Two things must both be true: the key track must be **routed/sent to the track hosting VisualComp 2**, and the plugin must be told to **listen to that external input**. Doing only one of them is the usual reason sidechain "does nothing".

FL STUDIO

1. Put VisualComp 2.25 on the mixer track you want ducked (for example, the bass).
2. Select the **kick's mixer track**, and at the bottom of the bass track's channel click the small send arrow so the kick is routed to the bass track. The arrow lights up to show the side–send is active.
3. Open the VisualComp 2.25 plugin window and click the **wrapper menu** (the arrow in the plugin's title bar).
4. Go to **Processing** → **Connections**, find the input labelled **External input** (or the sidechain pair) in the dropdown, and select the routed source.
5. Press **SC** in the plugin. The `SC DUCKING` stamp appears on the input display.

ABLETON LIVE

1. Place VisualComp 2.25 on the target track.
2. Open the plugin's device panel and set the **Sidechain** input to the key track using Live's plugin sidechain routing selector.
3. Press **SC**.

CUBASE / NUENDO

1. Insert the plugin, then activate its **sidechain** button in the plugin header.
2. On the key track, add a **send** targeting the plugin's sidechain input.
3. Press **SC**.

REAPER

1. Send the key track to the target track, setting the send's destination channels to **3/4**.
2. Set the target track's channel count to 4, then in the plugin's **pin connector** map channels 3/4 to the plugin's sidechain inputs.
3. Press **SC**.

STUDIO ONE / LOGIC (AU BUILD)

1. Choose the key track from the **sidechain** selector in the plugin header.
2. Press **SC**.

READING THE DUCKING OVERLAY

With SC engaged, a pale blue ceiling line and shaded region appear over the input waveform. **This is not the raw sidechain signal** – it is the actual gain reduction the compressor is applying, computed through your threshold, ratio, knee, attack and release. The further the line pushes down into the waveform, the deeper the duck at that instant.

- The line staying flat at the top means the key signal never crosses the threshold – no ducking is happening. Lower the threshold.
- A slow **attack** shows as a visible delay before the line dives, so you can see the kick's own transient escaping the duck.
- The **release** tail is the slope as the line climbs back up. Match its length to the space between beats and the groove locks in.
- A deep, narrow dip is a tight rhythmic duck; a broad shallow dip is gentle level–sharing.

CLASSIC EDM PUMP

Load the **EDM Pump** preset, engage SC, and adjust **RELEASE** until the recovery slope on the overlay finishes just as the next kick arrives. You are dialling groove by eye and confirming it by ear.

GOOD TO KNOW

The sidechain envelope is captured continuously whether or not SC is engaged, so switching the button on never shows stale data – the overlay is correct from the first instant.

11 Metering

Figure 11.1 – The metering column: needle meter above, transfer curve below.

GAIN REDUCTION METER

A ballistic needle showing gain reduction from 0 to −20 dB, with an orange arc tracing the current value. The scale is linear in decibels, so the needle always points at the true figure. Movement is deliberately weighted like a real meter: fast on attack, slower on release, so you read the compressor's character and not just a number.

The numeric read–out below gives the exact value. The **ACTIVE** lamp lights whenever more than 0.3 dB of reduction is occurring – an instant answer to "is this thing even doing anything?"

READING	APPLICATION
1 – 3 dB	Mastering, mix bus glue
2 – 5 dB	Drum bus, instrument buses
3 – 8 dB	Vocals, bass, individual tracks
10 dB +	Parallel compression, deliberate effect

TRANSFER CURVE

A live plot of input level (horizontal) against output level (vertical), both −60 to 0 dBFS:

- The **dotted diagonal** is unity gain – where the curve follows it, nothing is being changed.
- The **bend** is your threshold; the **slope after the bend** is your ratio. A shallower slope means firmer compression.
- The **shaded band** shows the knee width, and the **vertical red line** marks the threshold exactly.
- The **white dot** is the live operating point. Watching where it spends its time tells you whether your threshold is sensibly placed for the material.
- Current attack and release values are printed beneath the title for reference.

12 Gain Staging, Auto Gain & the Limiter

GAIN IN

±24 dB – applied before the detector. Raising it drives the signal further past the threshold, producing more compression without moving the threshold itself. This is how you push harder into a circuit to get more of its character, particularly in FET and Tube modes.

GAIN OUT

±24 dB – the final output level, applied after processing. Use it to restore the loudness that compression removed (traditionally called makeup gain).

AUTO GAIN

Continuously measures the input and post–compression signal levels and applies matching makeup gain automatically, up to 24 dB, smoothed over roughly 300 ms so it never audibly jumps.

Its real value is **honest comparison**. Louder almost always sounds better, which makes judging compression by ear notoriously unreliable. With Auto Gain engaged, bypassing the plugin compares processed and unprocessed signal at matched loudness – so you hear what the compressor actually did, not simply that it got louder.

> **IMPLEMENTATION NOTE**
> Output level is measured **before** auto–gain is applied, which prevents the feedback loop that would otherwise make the correction oscillate. Turning Auto Gain off resets its internal measurement immediately.

LIM – 0 DB CEILING

A brickwall limiter on the output, fixed at 0 dBFS. It is a safety net, not a mastering limiter: leave it on to guarantee nothing clips downstream. If it is working hard, reduce **GAIN OUT** rather than relying on it for level.

A PRACTICAL GAIN–STAGING ROUTINE

1. Start with both faders at 0 dB and note your normal playback level.
2. Set threshold and ratio for the gain reduction you want on the needle.
3. Engage **AUTO GAIN** and A/B against **BYPASS**. If the compressed version does not clearly sound better, your settings need work – not more level.
4. When satisfied, either leave Auto Gain engaged or dial the same amount manually into **GAIN OUT** for a fully static setting.
5. Enable **LIM** if the material needs a hard ceiling.

13 Preset Library

Thirty–four factory presets, each setting all nine parameters plus the detector circuit, so every one is a complete, usable starting point. Values follow long–established practice with classic bus compressors, FET and optical units, and modern mastering workflows.

MASTERING & MIX BUS

PRESET	THR	KNEE	RATIO	ATK	REL	MODE	NOTES
Mastering Glue (default)	-10	6	2:1	30	100	VCA	Transparent bus glue
Mastering Warm Tube	-8	8	1.5:1	50	300	Tube	Weight and warmth
Mastering Loud & Proud	-14	4	3:1	10	100	VCA	Limiter engaged
Mix Bus Punch	-12	3	4:1	30	100	VCA	Transient–forward
Mix Bus Smooth	-10	10	2:1	30	300	VCA	Wide knee, gentle
Gentle Tape Feel	-6	12	1.5:1	80	400	Tube	Barely–there softening

DRUMS

PRESET	THR	KNEE	RATIO	ATK	REL	MODE	NOTES
Drum Bus Glue	-12	4	4:1	30	150	VCA	The everyday drum bus
Drum Bus Smash	-30	2	10:1	1	100	FET	Mix at 35 % – parallel
Kick Punch	-15	2	4:1	30	100	FET	Slow attack keeps the thump
Snare Crack	-15	2	4:1	10	150	FET	Body without losing the hit
Room Crush	-35	0	12:1	0.5	80	FET	Extreme; blend to taste
Overheads Tame	-18	8	3:1	30	300	Optical	Controls cymbal wash
Percussion Control	-18	4	4:1	15	120	VCA	Evens erratic hits

Preset Library (continued)

VOCALS

PRESET	THR	KNEE	RATIO	ATK	REL	MODE	NOTES
Lead Vocal Smooth	-18	8	3:1	30	500	Optical	Effortless levelling
Lead Vocal Upfront	-20	4	4:1	5	200	FET	Presence and forwardness
Vocal Thickener	-16	6	3:1	50	300	Tube	Adds body and warmth
Backing Vocals Tight	-22	6	5:1	10	200	VCA	Sits stacks back
Rap Vocal In–Your–Face	-24	2	6:1	2	150	FET	Aggressive and constant
Vocal Leveller	-14	10	2.5:1	40	600	Optical	First stage of two
Whisper Boost	-30	8	4:1	20	400	Optical	Rescues quiet delivery

BASS

PRESET	THR	KNEE	RATIO	ATK	REL	MODE	NOTES
Bass Guitar Solid	-18	4	5:1	7	600	FET	Consistent note–to–note
Bass Synth Flat	-16	2	8:1	5	200	VCA	Rock–steady low end
Upright Warmth	-14	8	3:1	40	300	Tube	Keeps acoustic character
808 Control	-12	4	4:1	30	250	VCA	Tames sub weight

GUITAR & KEYS

PRESET	THR	KNEE	RATIO	ATK	REL	MODE	NOTES
Acoustic Guitar Even	-16	8	3:1	15	250	Optical	Smooths strumming
Electric Rhythm Tight	-18	4	4:1	5	200	VCA	Locks in the part
Funk Guitar Snap	-20	2	6:1	8	120	FET	Emphasises the pluck
Piano Glue	-14	8	2.5:1	30	300	VCA	Gentle, keeps dynamics
Synth Pad Smooth	-16	10	2.5:1	50	400	Optical	Levels evolving textures

Preset Library (continued)

UTILITY & FX

PRESET	THR	KNEE	RATIO	ATK	REL	MODE	NOTES
EDM Pump	-25	2	8:1	1	200	**VCA**	**Engage SC for ducking**
Podcast Levelling	-20	10	3:1	20	400	**Optical**	**Speech; limiter on**
Parallel Thickener	-28	4	8:1	3	150	**FET**	**Mix at 40 %**
Brickwall Safety	-6	2	10:1	0.5	80	**VCA**	**Peak catcher; limiter on**
Do Nothing (Reset)	0	0	1:1	30	100	**VCA**	**Neutral starting point**

Threshold in dB / Knee in dB / Attack and Release in milliseconds

14 Saving Your Own Presets

SAVING

1. Dial in a sound you want to keep.
2. Click **SAVE** in the preset strip (or choose **Save Preset As…** from the preset menu).
3. Name the preset and confirm. It is written as a `.vcpreset` file.

The saved name appears in the preset field, and your preset is listed under **MY PRESETS** at the bottom of the browser from then on.

WHAT GETS STORED

All nine parameters – threshold, knee, ratio, attack, release, input gain, output gain, mix and limiter state – plus the detector circuit.

WHERE THEY LIVE

WINDOWS	`Documents\Azazel Audio\VisualComp 2\Presets\`
MACOS	`~/Documents/Azazel Audio/VisualComp 2/Presets/`

Choose **Open Presets Folder** from the preset menu to jump straight there. The files are plain XML, so you can back them up, rename them, or share them with collaborators by copying them into the same folder on another machine.

> **TIP**
> Prefix related presets with a common word – `MyVox Verse`, `MyVox Chorus` – and they sort together in the browser.

PROJECT STATE

Independently of the preset system, your full settings, circuit mode, preset name and zoom level are saved inside your DAW project and restored exactly when you reopen it.

15 Recipes

MIX BUS GLUE

Load Mastering Glue. Lower the threshold until the needle reads 1—3 dB on the loudest passages. If it feels stiff, widen the knee or lengthen the release. The output waveform should look like the input with the peaks gently rounded – not flattened.

PUNCHY DRUM BUS

Drum Bus Glue in VCA, attack 20—30 ms so the stick attack escapes before compression bites, release around 150 ms so the compressor recovers between hits. Aim for 3—5 dB of reduction. If the kick is dominating the detector, raise the threshold slightly and accept less reduction.

VOCAL THAT SITS WITHOUT EFFORT

Two gentle stages beat one aggressive one. Start with Vocal Leveller (Optical, 2.5:1, slow) to even out the performance, then a second instance with Lead Vocal Upfront (FET, 4:1, fast) for presence. Each doing 3—4 dB is far more natural than one doing 8 dB.

PARALLEL (NEW YORK) COMPRESSION

Load Parallel Thickener or Drum Bus Smash. These crush deliberately – 10:1, very fast attack – then sit at 35—40 % mix. The compressed layer raises the quiet detail while the dry signal preserves every transient. Increase the mix until it feels solid, then back off slightly.

KICK–DUCKED BASS

Route the kick to the sidechain input (section 10), load EDM Pump, engage **SC**. Watch the ducking overlay: the dip should begin exactly on the kick and recover just before the next one. Threshold sets depth, release sets groove.

TAMING AN UNPREDICTABLE PERFORMANCE

Whisper Boost or Vocal Leveller in Optical mode, with a long release. Engage **AUTO GAIN** so the quiet passages come up to meet the loud ones, and add **LIM** to stop the occasional shout from clipping.

LEARNING WHAT ATTACK REALLY DOES

On a drum loop, set ratio 6:1 and threshold for 6 dB of reduction. Now sweep the attack from 0.1 ms to 100 ms while watching the output display – hold **Ctrl** for a slow, precise sweep. You will see the transients reappear as the attack lengthens, and hear the loop change from soft to punchy. This single exercise teaches more than any explanation.

16 Troubleshooting

SYMPTOM	CAUSE AND REMEDY
Plugin does not appear in the DAW	Confirm the `.vst3` is in the standard VST3 folder (section 02) and rescan. Custom search paths do not apply to VST3 in FL Studio. On macOS, clear the quarantine flag.
Updated the plugin but nothing changed	DAWs keep the loaded binary in memory. Remove the plugin instance (or close the project), then re-add it.
No gain reduction; needle never moves	Signal is not reaching the threshold. Lower **THRESHOLD** or raise **GAIN IN**. Check **BYPASS** is off and, if **SC** is engaged, that the sidechain bus is actually receiving audio.
Sidechain does nothing	Both halves of the routing must be done: the key track sent to this track, and the plugin's external input selected (section 10). In FL Studio the wrapper path is Processing → Connections.
Compression sounds harsh or grainy	Attack and release are too fast for the material, especially on bass content. Lengthen both, widen the **KNEE**, or switch to Optical or Tube mode.
Audible pumping or breathing	Release is fighting the programme. Lengthen it, reduce the ratio, or lower **MIX** so dry signal masks the movement.
Everything sounds lifeless	Over-compression. Raise the threshold, lengthen the attack to restore transients, and A/B with **AUTO GAIN** engaged so loudness is not fooling you.
Hard to set an exact value	Hold **Ctrl** (Cmd on macOS) while dragging for fine adjustment, or double-click to reset and start again.
Output clipping downstream	Reduce **GAIN OUT** and enable **LIM** for a hard 0 dB ceiling.
Interface too large or too small	Use the zoom selector in the header (50 %—200 %). The setting persists with the project.
Want the tour again	Click the Azazel logo in the top-left corner and choose **Show Help**.
A saved preset is missing	Choose **Open Presets Folder** from the preset menu and confirm the `.vcpreset` file is present. Files must sit directly in that folder, not in subfolders.

STILL STUCK?

Click the Azazel logo and choose **Visit azazelaudio.com**.

17 Version 2.1 – New Tools

Version 2.1 adds a parametric EQ, an output clipping stage, level metering, an Auto–Analyze preset wizard, and two small interface refinements. This section documents each one, including the relocated zoom control covered in item 17.1 below.

17.1 – THE INVISIBLE ZOOM CONTROL

The header no longer carries a separate, visible zoom button. Instead, **zoom now lives at the very top of the Azazel logo menu** – click the logo (top–left corner; it has no visible border or highlight of its own, but the cursor changes to a hand when it is hovered) and the first entry is **Zoom** (current %), opening the same 50—200% choices as before. This keeps the header uncluttered: the logo is the single interactive mark there, and everything else – zoom, Show Help, and the link to azazelaudio.com – is one click behind it.

> **WHY "INVISIBLE"**
>
> The logo hot–spot deliberately draws no highlight, outline or background tint at rest or on hover – only the pointing–hand cursor gives away that it is clickable. This was a specific design request: the mark should look like branding, not a button, until the moment you actually use it.

17.2 – PARAMETRIC EQ

Figure 17.1 – The docked EQ panel, opened with the new **EQ** button (left of the Mode selector). It extends to the left of the main interface at full height and does not resize or crop anything else.

Up to 8 nodes, each independently a Bell, Low Shelf, High Shelf, High Pass or Low Pass filter:

- **Click empty graph space** to activate the next free node at that frequency and gain.
- **Drag a node** – left/right changes frequency, up/down changes gain.
- **Mouse wheel** over a node adjusts its Q (bandwidth).
- **Right–click** a node for its menu: filter type, link to the compressor, or remove it.

Each node is numbered and colour–coded (1—8); the row list below the graph shows every node's type, frequency and gain, with a **LINK** toggle per row – the same colour and number appear in the Dynamics pane (17.3) when that node is the one currently driving the compressor.

17.3 – MULTIBAND–AWARE DETECTION

Any EQ node marked **LINK** feeds an independent band–energy detector into the compressor, in addition to the normal broadband level – so a hot linked band can trigger compression even when the overall signal is under threshold. The small numbered box at the top–left of the Dynamics pane (where the knob labels used to start; they are now one row lower) shows which linked node is currently dominant, in that node's colour, or a dash when none is active.

> **WHAT THIS IS, AND ISN'T**
>
> This is a single compressor whose detector can be made sensitive to specific frequency bands you choose in the EQ – not eight independent per–band compressors summed back together. It is the simpler, more transparent design, and it is what the on–screen "one active band" indicator actually shows.

17.4 – OUTPUT CLIPPING MODES

A button beside Bypass (labelled with the current mode) cycles the final output stage through three options, all sharing one fixed 3 ms lookahead so switching between them live never causes a click or a latency jump:

MODE	BEHAVIOUR	USE FOR
Soft Clip	tanh–style saturation with an audible, deliberate character as it nears 0 dBFS	Colour, glue, drum/parallel crush
Brickwall	lookahead limiter, transparent, hard 0 dB ceiling	Mastering, safety on any loud bus
Off	no processing at all	Serial/gain–staging presets that must stay clean

All 34 factory presets now set a clip mode that matches their intent – mastering and "loud" presets default to Brickwall; tape/tube/parallel–crush presets default to Soft; clean gain–staging presets default to Off.

17.5 – LEVEL METERING

Figure 17.2 – The narrow strip at the far right of the GR meter / transfer curve column is the new level meter.

A colour–coded vertical dB meter (green/yellow/red, peak + RMS) sits at the right edge of the meter column. **Click** it to slide out a second bar showing an approximate LUFS reading (momentary and short–term). This is a K–weighting–style approximation using continuous smoothing rather than the discrete gated blocks of the ITU–R BS.1770 standard – close enough to gauge loudness while mixing, but not a certified delivery–spec measurement.

17.6 – AUTO–ANALYZE WIZARD

Next to the preset controls, the **AUTO** button starts a two–step wizard:

1. **Genre** – choose the closest match from Pop, Hip–Hop / Trap, EDM / Dance, Rock / Metal, Acoustic / Folk, Classical / Orchestral, or Podcast / Spoken Word.
2. **Target LUFS** – a numeric field, pre–filled with a sensible default for the chosen genre.

On confirmation, it measures the crest factor and level of the material already in the plugin's input buffer, chooses ratio/attack/release/circuit/ clip–mode from a per–genre character table, sets the threshold at the midpoint between the measured peak and RMS, and trims output gain toward your target using the plugin's own LUFS estimate – then applies the result as a new preset named **"Auto: <genre> <LUFS> LU"**.

> **WHAT AUTO–ANALYZE IS NOT**
>
> This is a heuristic starting point built from a real measurement of your audio, not a trained mastering AI. Treat the result the same as any factory preset: a considered starting point to refine by ear, not a finished master.

17.7 – REACTIVE HEADER GLOW

The orange accent line beneath the plugin name idles with a slow random shimmer and flares briefly whenever the plugin window is dragged to a new position on screen – a small piece of visual polish with no effect on audio or parameters.

17.8 – REBRANDED MARK

The Azazel logo now shows the wordmark alone; the "Audio" script that previously sat beneath it has been removed for a cleaner, more compact mark at small sizes.

18 Version 2.2 – Multiband Dynamic EQ

Version 2.2 turns the parametric EQ from Section 17.2 into a real per–band dynamics tool. Any node can now carry its own Threshold, Knee, Ratio, Range, Attack and Release – independent of every other node's – layered on top of the single broadband compressor described in Sections 08—09. This section covers what changed; the control layout itself (Threshold/Knee/Ratio/ Attack/Release knobs, the numbered band–selector row) is unchanged from Section 08.

18.1 – MULTIBAND IS ALWAYS ON

The old **MULTIBAND COMP** toggle button is gone. Multiband dynamics are permanently active – every linked node always runs its own detector and dynamics – and this is not saved as a per–project setting, so there is nothing to remember to turn back on. The EQ panel's open/close button (top–left of the header strip) is now labelled **MB** instead of **EQ**, but does the same thing: it only opens or closes the docked panel.

> **WHAT "MULTIBAND" MEANS HERE**
>
> This is still not eight fully independent compressors summed together – there is one broadband compressor underneath, and each linked node additionally runs its own dynamic–EQ–style gain adjustment at its own frequency. That combination is what every control in this chapter edits.

18.2 – PER–BAND DYNAMICS & THE THRESHOLD TAPER

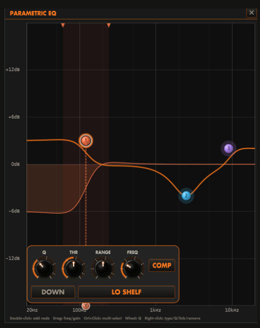

Figure 18.1 – Node 1's own Q/Threshold/Range/Freq controls in the Dynamic Island, mirrored below by the main Threshold/Knee/Ratio/Attack/Release knobs.

Select a node (its number lights up below the graph), then use the Threshold/Knee/Ratio/Attack/Release knobs from Section 08 – they now edit that node instead of the shared broadband compressor:

CONTROL	RANGE	DEFAULT
Threshold	`0 to -60 dB`	`-20 dB`
Knee	`0 to 20 dB`	`6 dB`
Ratio	`1:1 to 20:1`	`2:1`
Attack	`0.1 to 200 ms`	`0.2 ms`
Release	`1 to 2000 ms`	`45 ms`

The Threshold knob has a deliberately uneven response: turning it uses a logarithmic taper so the knob's **halfway point lands on −20 dB** – a musically useful "sweet spot" – rather than the midpoint of the raw 0 to −60 dB range. That gives finer control across the −20 to 0 dB range most material actually sits in, and compresses the rarely−used −60 to −40 dB tail into the remaining half−turn. The same mapping, and the same underlying value, drives the draggable orange **T** marker on the graph itself (visible in Figure 18.1, below node 1).

18.3 – UPWARD VS. DOWNWARD, VIA RANGE

The **Range** knob (Dynamic Island only, Ó30 dB) sets both how far this band's dynamics can move its gain and, through its sign, which direction they move it:

- **Negative Range** (the default) – ordinary downward compression: material above Threshold is turned down, up to Range's magnitude.
- **Positive Range** – upward compression: material below Threshold is turned up instead, using the same Knee/Ratio shape mirrored around the threshold. Useful for lifting buried detail (room tone, a sustain tail, a quiet vocal phrase) without touching anything already loud.

The Dynamic Island's **UP** / **DOWN** button reflects and can override this, but turning Range past zero in either direction will flip it automatically – you rarely need to touch the button directly.

18.4 – THE DYNAMIC ISLAND

Clicking any node opens a small floating control popup docked to its spot on the graph – the **Dynamic Island**. Alongside Q (also reachable by mouse wheel over the node, or the right–click Q submenu) and the Threshold/Range pair from 18.2—18.3, it carries two more controls:

- **FREQ** – the node's own frequency (20 Hz to 20 kHz). Identical to dragging the node left/right on the graph, but useful for a precise numeric–feeling nudge without also disturbing gain.
- **COMP** – links or unlinks this node from the compressor. A newly created node (double–click on empty graph space) is **linked by default**, so a new band gets its own dynamics immediately rather than needing a second step.

Every Island edit, and every graph drag, updates the Dynamics pane's knobs and progress bars immediately – there is no need to click elsewhere or wait for the display to refresh.

18.5 – MULTI–SELECT EDITING

Ctrl+Click (Cmd+Click on macOS) a second node to add it to a selection; the most recently clicked node becomes the anchor. Dragging the anchor's frequency/gain, or turning its Q, now applies the same relative change to every other selected node – a dB delta for gain, a multiplicative ratio for frequency and Q – each clamped to its own valid range. A plain click (no modifier) always resets the selection back to just that one node. Right–click actions (change type, link/unlink, remove) always apply to the single clicked node regardless of any active multi–selection.

> **DETECTOR EDGES, BRIEFLY**
>
> Each linked node's shaded band strip (visible in Figure 18.1) still shows the frequency range feeding its detector, independent of its own tone shape – see Section 19.3 for how dragging those edges changed in 2.21.

19 Version 2.21 – Refinements

Version 2.21 is a smaller, interface–focused update on top of 2.2's multiband dynamic EQ: a more predictable home for the Dynamic Island, a clearer level meter, and detector–edge dragging that is both easier to grab and easier to line bands up with.

19.1 – DYNAMIC ISLAND DOCKED TO THE GRAPH

The Dynamic Island no longer hovers above or below whichever node is selected, flipping sides depending on how close that node is to the top or bottom of the graph. It now docks to a fixed spot at the bottom of the EQ graph, 5 px above the frequency axis, sliding only left/right to stay centred under the selected node (Figure 18.1 shows this docked position). The result is one predictable location to look for it regardless of which node – or how high or low on the graph – is selected.

19.2 – LEVEL METER: PEAK–HOLD AND SIDE LABELS

Figure 19.1 – The level meter strip (far right edge, shown here next to the Gain Reduction meter for scale), both bars revealed. The readout at top is the peak–hold value; "dB" and "LUFS" now label their bars from the side.

The dB and LUFS unit labels introduced in Section 11 have moved from underneath each bar to a narrow vertical label immediately to its right, in bright white for quick reading at a glance. That freed the space they used to occupy, which now goes to a numeric peak–hold readout above the bars: the highest dB peak reached in the trailing three seconds, rather than an ordinary meter needle that can slip back down the instant a transient passes. It reads "––" when there has been no signal in that window.

19.3 – CLICK–TO–JUMP BORDERS & PERSISTENT SNAP JUNCTIONS

Every linked node's detector–edge border (the shaded band's low/high boundary, Section 18.5) can now be grabbed by clicking **anywhere near it** along the graph's full height, not just the small flag triangle at the very top – and a single click, without needing to drag first, jumps that border straight to the clicked frequency.

Borders dragged within 100 Hz of another linked node's edge still snap together, as introduced in 2.2 – but that snap is now a lasting bond, not a one–time nudge:

- **Dragging a snapped junction** moves both nodes' edges together, keeping them locked to the same frequency as you drag left or right.
- **Moving a node itself** (dragging its centre frequency) carries any junction it participates in along with it, so the bond survives instead of drifting apart the next time that node moves.
- **Newly created nodes** snap their default edges to any nearby linked node right away, so placing a band next to an existing one lines up the crossover without any manual nudging.

Dragging a border far enough away from its partner (out past the 100 Hz snap radius) breaks the bond, freeing that edge to move independently again.

20 Specifications & Credits

PARAMETERS

PARAMETER	RANGE	DEFAULT
Threshold	-60 to 0 dB	-10 dB
Knee	0 to 20 dB	6 dB
Ratio	1:1 to 20:1	2:1
Attack	0.1 to 200 ms	30 ms
Release	1 to 2000 ms	100 ms
Gain In / Gain Out	-24 to +24 dB	0 dB
Mix	0 to 100 %	100 %
Limiter	Off / 0 dB ceiling	Off
Detector circuit	VCA / FET / Optical / Tube	VCA

TECHNICAL

- Formats: VST3 and Standalone application
- Windows 10 / 11 (64-bit); macOS 10.13 or later, universal binary (Apple Silicon + Intel)
- Channels: mono or stereo main bus, plus an optional stereo sidechain bus
- Sample-accurate envelope detection with per-circuit ballistics
- Soft-knee transfer function with fully variable knee width
- Lock-free visualisation pipeline – the audio thread never waits on the interface
- Auto-gain measured pre-makeup to avoid feedback artefacts
- Full parameter automation; complete state saved with the host project
- Interface rendered as vector graphics at every zoom level

CREDITS

VisualComp 2.25 was designed, programmed and voiced by **Arnav Singh for Azazel Audio**. Built with the JUCE framework.

Preset values were informed by long-documented practice around the SSL G-Series bus compressor, FET and optical studio units, and contemporary mastering technique – then verified by ear on real material.

> **THANK YOU**
>
> Thank you for using VisualComp 2.25. If it helps you understand compression a little better, it has done its job. – azazelaudio.com